

Literature Review Workflow Library

Expert Workflow Library

Ultimate AI Research Toolkit 2026 - Premium Edition
Designed for PhD students, professors, graduate researchers, engineers, and data scientists.

[Idea]	Research prompts and ideation systems
[Review]	Literature synthesis and gap detection
[Write]	Paper writing and revision workflows
[Plan]	Publication and productivity operations

Table of Contents

1. Introduction
2. Instructions
3. Workflow Library
4. Summary

Introduction

This Literature Review collection provides 300 expert research workflows engineered for high-rigor academic and technical delivery. Each workflow includes practical fields to help users move from strategy to publication-ready output.

Instructions

- Select workflows by category and difficulty to match project maturity.
- Replace variables in each workflow with your discipline, data, and resource constraints.
- Use Expected Output as the acceptance criteria for quality control.
- Store outputs in versioned folders to preserve decision history.

Workflow Library

1. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - practical deployment framing

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize rigor and produce publication-grade decisions for Critical Appraisal.

Prompt: Rewrite the methodology around Critical Appraisal for state-of-the-art positioning for rebuttal so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ACM CSUR. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=benchmarking specialist. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Strategy

2. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - method comparison clarity

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize reproducibility and produce publication-grade decisions for Gap Identification.

Prompt: Write a Nature-style abstract for a study on Gap Identification in rapid scoping review for proposal. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=theory-oriented reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Analysis

3. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - editorial risk reduction

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize clarity and produce publication-grade decisions for Synthesis Design.

Prompt: Generate a figure strategy for Synthesis Design with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Knowledge-Based Systems. Add one caption rewrite for non-expert readers. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=methodology purist. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Writing

4. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - venue-specific optimization

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize statistical power and produce publication-grade decisions for Search Strategy.

Prompt: Build an ethics checklist and mitigation plan for Search Strategy in systematic review planning phase. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for PRISMA-aligned journal review. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=interdisciplinary committee member. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Publication

5. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - decision traceability

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize transferability and produce publication-grade decisions for Critical Appraisal.

Prompt: Design a publication strategy for Critical Appraisal: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ACM CSUR. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=ethics compliance reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Methodology

6. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - citation-strength positioning

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize ethical compliance and produce publication-grade decisions for Gap Identification.

Prompt: Assume Reviewer #2 challenges Gap Identification for IEEE Survey Journal. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=reproducibility auditor. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Execution

7. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - defense-ready

reasoning

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize computational efficiency and produce publication-grade decisions for Synthesis Design.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Synthesis Design. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for Elsevier Knowledge-Based Systems. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=applied domain expert. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Compliance

8. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - high-confidence statistical reporting

Objective: Select robust statistical strategy and interpretation logic. Prioritize clinical relevance and produce publication-grade decisions for Search Strategy.

Prompt: Design a statistical analysis workflow for Search Strategy with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for PRISMA-aligned journal. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=systems performance reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Reasoning

9. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - evidence-first writing

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize publication readiness and produce publication-grade decisions for Critical Appraisal.

Prompt: Analyze a draft on Critical Appraisal and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to ACM CSUR. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=industry impact reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Analysis

10. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - hypothesis stress testing

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize novelty and produce publication-grade decisions for Gap Identification.

Prompt: Interpret results for Gap Identification using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in IEEE Survey Journal. Add one paragraph that preempts overclaiming. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=editor focused on clarity. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Writing

11. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - results-to-insight translation

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize rigor and produce publication-grade decisions for Synthesis Design.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Synthesis Design and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for Elsevier Knowledge-Based Systems. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=skeptical statistician. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Publication

12. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - major-revision readiness

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize reproducibility and produce publication-grade decisions for Search Strategy.

Prompt: Generate a publication-ready section for Search Strategy tuned for PRISMA-aligned journal. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=benchmarking specialist. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Methodology

13. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - replication-grade

documentation

Objective: Pressure-test experimental design before expensive execution. Prioritize clarity and produce publication-grade decisions for Critical Appraisal.

Prompt: Audit the experimental design for Critical Appraisal. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for ACM CSUR reviewers. Include a stop/go decision protocol. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=theory-oriented reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Execution

14. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - reviewer-resilient argumentation

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize statistical power and produce publication-grade decisions for Gap Identification.

Prompt: Create a LaTeX writing workflow for Gap Identification: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for IEEE Survey Journal submission checks. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=methodology purist. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Compliance

15. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - ethics-by-design reporting

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize transferability and produce publication-grade decisions for Synthesis Design.

Prompt: Simulate a high-pressure Q&A; on Synthesis Design. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=interdisciplinary committee member. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Reasoning

16. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - limitations-first transparency

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize ethical compliance and produce publication-grade decisions for Search Strategy.

Prompt: Act as a senior research editor. Build a contribution architecture for Search Strategy in systematic review planning phase targeting PRISMA-aligned journal. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=ethics compliance reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Rebuttal

17. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - cross-domain readability

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize computational efficiency and produce publication-grade decisions for Critical Appraisal.

Prompt: Rewrite the methodology around Critical Appraisal for state-of-the-art positioning for rebuttal so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ACM CSUR. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=reproducibility auditor. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Validation

18. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - impact-focused communication

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize clinical relevance and produce publication-grade decisions for Gap Identification.

Prompt: Write a Nature-style abstract for a study on Gap Identification in rapid scoping review for proposal. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=applied domain expert. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Publication

19. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - publishable novelty framing

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize publication readiness and produce publication-grade decisions for Synthesis Design.

Prompt: Generate a figure strategy for Synthesis Design with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Knowledge-Based Systems. Add one caption rewrite for non-expert readers. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=systems performance reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Methodology

20. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - practical deployment framing

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize novelty and produce publication-grade decisions for Search Strategy.

Prompt: Build an ethics checklist and mitigation plan for Search Strategy in systematic review planning phase. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for PRISMA-aligned journal review. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=industry impact reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Execution

21. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - method comparison clarity

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize rigor and produce publication-grade decisions for Critical Appraisal.

Prompt: Design a publication strategy for Critical Appraisal: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ACM CSUR. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=editor focused on clarity. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Compliance

22. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - editorial risk reduction

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize reproducibility and produce publication-grade decisions for Gap Identification.

Prompt: Assume Reviewer #2 challenges Gap Identification for IEEE Survey Journal. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=skeptical statistician. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Reasoning

23. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - venue-specific optimization

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize clarity and produce publication-grade decisions for Synthesis Design.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Synthesis Design. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure

that anticipates editor priorities for Elsevier Knowledge-Based Systems. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=benchmarking specialist. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Rebuttal

24. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - decision traceability

Objective: Select robust statistical strategy and interpretation logic. Prioritize statistical power and produce publication-grade decisions for Search Strategy.

Prompt: Design a statistical analysis workflow for Search Strategy with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for PRISMA-aligned journal. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=theory-oriented reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Validation

25. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - citation-strength positioning

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize transferability and produce publication-grade decisions for Critical Appraisal.

Prompt: Analyze a draft on Critical Appraisal and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to ACM CSUR. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=methodology purist; output

schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=methodology purist. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Optimization

26. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - defense-ready reasoning

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize ethical compliance and produce publication-grade decisions for Gap Identification.

Prompt: Interpret results for Gap Identification using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in IEEE Survey Journal. Add one paragraph that preempts overclaiming. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=interdisciplinary committee member. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Planning

27. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - high-confidence statistical reporting

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize computational efficiency and produce publication-grade decisions for Synthesis Design.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Synthesis Design and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for Elsevier Knowledge-Based Systems. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as

numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=ethics compliance reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Execution

28. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - evidence-first writing

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize clinical relevance and produce publication-grade decisions for Search Strategy.

Prompt: Generate a publication-ready section for Search Strategy tuned for PRISMA-aligned journal. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=reproducibility auditor. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Compliance

29. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - hypothesis stress testing

Objective: Pressure-test experimental design before expensive execution. Prioritize publication readiness and produce publication-grade decisions for Critical Appraisal.

Prompt: Audit the experimental design for Critical Appraisal. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for ACM CSUR reviewers. Include a stop/go decision protocol. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=applied domain expert. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Reasoning

30. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - results-to-insight translation

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize novelty and produce publication-grade decisions for Gap Identification.

Prompt: Create a LaTeX writing workflow for Gap Identification: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for IEEE Survey Journal submission checks. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=systems performance reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Rebuttal

31. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - major-revision readiness

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize rigor and produce publication-grade decisions for Synthesis Design.

Prompt: Simulate a high-pressure Q&A; on Synthesis Design. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=industry impact reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Validation

32. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - replication-grade documentation

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize reproducibility and produce publication-grade decisions for Search Strategy.

Prompt: Act as a senior research editor. Build a contribution architecture for Search Strategy in systematic review planning phase targeting PRISMA-aligned journal. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=editor focused on clarity. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Optimization

33. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - reviewer-resilient argumentation

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize clarity and produce publication-grade decisions for Critical Appraisal.

Prompt: Rewrite the methodology around Critical Appraisal for state-of-the-art positioning for rebuttal so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ACM CSUR. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=skeptical statistician. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Planning

34. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - ethics-by-design reporting

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize statistical power and produce publication-grade decisions for Gap Identification.

Prompt: Write a Nature-style abstract for a study on Gap Identification in rapid scoping review for proposal. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=benchmarking specialist. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Strategy

35. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - limitations-first transparency

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize transferability and produce publication-grade decisions for Synthesis Design.

Prompt: Generate a figure strategy for Synthesis Design with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Knowledge-Based Systems. Add one caption rewrite for non-expert readers. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=theory-oriented reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Analysis

36. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - cross-domain readability

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize ethical compliance and produce publication-grade decisions for Search Strategy.

Prompt: Build an ethics checklist and mitigation plan for Search Strategy in systematic review planning phase. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for PRISMA-aligned journal review. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=methodology purist. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Reasoning

37. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - impact-focused communication

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize computational efficiency and produce publication-grade decisions for Critical Appraisal.

Prompt: Design a publication strategy for Critical Appraisal: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ACM CSUR. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=interdisciplinary committee member. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Rebuttal

38. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - publishable novelty framing

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize clinical relevance and produce publication-grade decisions for Gap Identification.

Prompt: Assume Reviewer #2 challenges Gap Identification for IEEE Survey Journal. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=ethics compliance reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Validation

39. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - practical deployment framing

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize publication readiness and produce publication-grade decisions for Synthesis Design.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Synthesis Design. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for Elsevier Knowledge-Based Systems. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=reproducibility auditor. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Optimization

40. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - method comparison clarity

Objective: Select robust statistical strategy and interpretation logic. Prioritize novelty and produce publication-grade decisions for Search Strategy.

Prompt: Design a statistical analysis workflow for Search Strategy with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for PRISMA-aligned journal. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=applied domain expert. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Planning

41. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - editorial risk reduction

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize rigor and produce publication-grade decisions for Critical Appraisal.

Prompt: Analyze a draft on Critical Appraisal and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to ACM CSUR. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=systems performance reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Strategy

42. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - venue-specific optimization

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize reproducibility and produce publication-grade decisions for Gap Identification.

Prompt: Interpret results for Gap Identification using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in IEEE Survey Journal. Add one paragraph that preempts overclaiming. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=industry impact reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Analysis

43. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - decision traceability

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize clarity and produce publication-grade decisions for Synthesis Design.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Synthesis Design and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for Elsevier Knowledge-Based Systems. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=editor focused on clarity. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Writing

44. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - citation-strength positioning

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize statistical power and produce publication-grade decisions for Search Strategy.

Prompt: Generate a publication-ready section for Search Strategy tuned for PRISMA-aligned journal. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=skeptical statistician. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Publication

45. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - defense-ready reasoning

Objective: Pressure-test experimental design before expensive execution. Prioritize transferability and produce publication-grade decisions for Critical Appraisal.

Prompt: Audit the experimental design for Critical Appraisal. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for ACM CSUR reviewers. Include a stop/go decision protocol. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=benchmarking specialist. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Validation

46. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - high-confidence statistical reporting

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize ethical compliance and produce publication-grade decisions for Gap Identification.

Prompt: Create a LaTeX writing workflow for Gap Identification: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for IEEE Survey Journal submission checks. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=theory-oriented reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Optimization

47. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - evidence-first writing

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize computational efficiency and produce publication-grade decisions for Synthesis Design.

Prompt: Simulate a high-pressure Q&A; on Synthesis Design. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=methodology purist. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Planning

48. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - hypothesis stress testing

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize clinical relevance and produce publication-grade decisions for Search Strategy.

Prompt: Act as a senior research editor. Build a contribution architecture for Search Strategy in systematic review planning phase targeting PRISMA-aligned journal. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=interdisciplinary committee member. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Strategy

49. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - results-to-insight translation

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize publication readiness and produce publication-grade decisions for Critical Appraisal.

Prompt: Rewrite the methodology around Critical Appraisal for state-of-the-art positioning for rebuttal so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ACM CSUR. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=ethics compliance reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Analysis

50. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - major-revision readiness

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize novelty and produce publication-grade decisions for Gap Identification.

Prompt: Write a Nature-style abstract for a study on Gap Identification in rapid scoping review for proposal. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=reproducibility auditor. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Writing

51. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - replication-grade documentation

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize rigor and produce publication-grade decisions for Synthesis Design.

Prompt: Generate a figure strategy for Synthesis Design with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Knowledge-Based Systems. Add one caption rewrite for non-expert readers. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=applied domain expert. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from an applied domain expert.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Publication

52. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - reviewer-resilient argumentation

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize reproducibility and produce publication-grade decisions for Search Strategy.

Prompt: Build an ethics checklist and mitigation plan for Search Strategy in systematic review planning phase. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for PRISMA-aligned journal review. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=systems performance reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Methodology

53. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - ethics-by-design reporting

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize clarity and produce publication-grade decisions for Critical Appraisal.

Prompt: Design a publication strategy for Critical Appraisal: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ACM CSUR. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=industry impact reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Execution

54. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - limitations-first transparency

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize statistical power and produce publication-grade decisions for Gap Identification.

Prompt: Assume Reviewer #2 challenges Gap Identification for IEEE Survey Journal. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods

protocol; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=editor focused on clarity. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Planning

55. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - cross-domain readability

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize transferability and produce publication-grade decisions for Synthesis Design.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Synthesis Design. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for Elsevier Knowledge-Based Systems. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=skeptical statistician. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Strategy

56. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - impact-focused communication

Objective: Select robust statistical strategy and interpretation logic. Prioritize ethical compliance and produce publication-grade decisions for Search Strategy.

Prompt: Design a statistical analysis workflow for Search Strategy with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for PRISMA-aligned journal. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=benchmarking

specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=benchmarking specialist. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Analysis

57. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - publishable novelty framing

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize computational efficiency and produce publication-grade decisions for Critical Appraisal.

Prompt: Analyze a draft on Critical Appraisal and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to ACM CSUR. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=theory-oriented reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Writing

58. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - practical deployment framing

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize clinical relevance and produce publication-grade decisions for Gap Identification.

Prompt: Interpret results for Gap Identification using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in IEEE Survey Journal. Add one paragraph that preempts overclaiming. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=methodology purist. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Publication

59. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - method comparison clarity

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize publication readiness and produce publication-grade decisions for Synthesis Design.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Synthesis Design and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for Elsevier Knowledge-Based Systems. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=interdisciplinary committee member. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Methodology

60. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - editorial risk reduction

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize novelty and produce publication-grade decisions for Search Strategy.

Prompt: Generate a publication-ready section for Search Strategy tuned for PRISMA-aligned journal. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=ethics compliance reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Execution

61. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - venue-specific optimization

Objective: Pressure-test experimental design before expensive execution. Prioritize rigor and produce publication-grade decisions for Critical Appraisal.

Prompt: Audit the experimental design for Critical Appraisal. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for ACM CSUR reviewers. Include a stop/go decision protocol. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=reproducibility auditor. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Compliance

62. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - decision traceability

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize reproducibility and produce publication-grade decisions for Gap Identification.

Prompt: Create a LaTeX writing workflow for Gap Identification: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for IEEE Survey Journal submission checks. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=applied domain expert. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Reasoning

63. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - citation-strength positioning

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize clarity and produce publication-grade decisions for Synthesis Design.

Prompt: Simulate a high-pressure Q&A; on Synthesis Design. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=systems performance reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Analysis

64. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - defense-ready reasoning

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize statistical power and produce publication-grade decisions for Search Strategy.

Prompt: Act as a senior research editor. Build a contribution architecture for Search Strategy in systematic review planning phase targeting PRISMA-aligned journal. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=industry impact reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact

reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Writing

65. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - high-confidence statistical reporting

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize transferability and produce publication-grade decisions for Critical Appraisal.

Prompt: Rewrite the methodology around Critical Appraisal for state-of-the-art positioning for rebuttal so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ACM CSUR. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=editor focused on clarity. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Publication

66. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - evidence-first writing

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize ethical compliance and produce publication-grade decisions for Gap Identification.

Prompt: Write a Nature-style abstract for a study on Gap Identification in rapid scoping review for proposal. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=skeptical statistician. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Methodology

67. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - hypothesis stress testing

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize computational efficiency and produce publication-grade decisions for Synthesis Design.

Prompt: Generate a figure strategy for Synthesis Design with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Knowledge-Based Systems. Add one caption rewrite for non-expert readers. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=benchmarking specialist. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Execution

68. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - results-to-insight translation

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize clinical relevance and produce publication-grade decisions for Search Strategy.

Prompt: Build an ethics checklist and mitigation plan for Search Strategy in systematic review planning phase. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for PRISMA-aligned journal review. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=theory-oriented reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Compliance

69. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - major-revision readiness

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize publication readiness and produce publication-grade decisions for Critical Appraisal.

Prompt: Design a publication strategy for Critical Appraisal: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ACM CSUR. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=methodology purist. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Reasoning

70. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - replication-grade documentation

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize novelty and produce publication-grade decisions for Gap Identification.

Prompt: Assume Reviewer #2 challenges Gap Identification for IEEE Survey Journal. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=interdisciplinary committee member. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Rebuttal

71. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - reviewer-resilient argumentation

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize rigor and produce publication-grade decisions for Synthesis Design.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Synthesis Design. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for Elsevier Knowledge-Based Systems. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=ethics compliance reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Validation

72. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - ethics-by-design reporting

Objective: Select robust statistical strategy and interpretation logic. Prioritize reproducibility and produce publication-grade decisions for Search Strategy.

Prompt: Design a statistical analysis workflow for Search Strategy with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for PRISMA-aligned journal. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=reproducibility auditor. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Publication

73. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - limitations-first transparency

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize clarity and produce publication-grade decisions for Critical Appraisal.

Prompt: Analyze a draft on Critical Appraisal and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to ACM CSUR. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=applied domain expert. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Methodology

74. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - cross-domain readability

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize statistical power and produce publication-grade decisions for Gap Identification.

Prompt: Interpret results for Gap Identification using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in IEEE Survey Journal. Add one paragraph that preempts overclaiming. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=systems performance reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Execution

75. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - impact-focused communication

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize transferability and produce publication-grade decisions for Synthesis Design.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Synthesis Design and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for Elsevier Knowledge-Based Systems. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=industry impact reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Compliance

76. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - publishable novelty framing

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize ethical compliance and produce publication-grade decisions for Search Strategy.

Prompt: Generate a publication-ready section for Search Strategy tuned for PRISMA-aligned journal. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=editor focused on clarity. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Reasoning

77. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - practical deployment framing - Variant 077

Objective: Pressure-test experimental design before expensive execution. Prioritize computational efficiency and produce publication-grade decisions for Critical Appraisal.

Prompt: Audit the experimental design for Critical Appraisal. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for ACM CSUR reviewers. Include a stop/go decision protocol. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=skeptical statistician. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Rebuttal

78. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - method comparison clarity - Variant 078

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize clinical relevance and produce publication-grade decisions for Gap Identification.

Prompt: Create a LaTeX writing workflow for Gap Identification: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for IEEE Survey Journal submission checks. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=benchmarking specialist. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Validation

79. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - editorial risk reduction - Variant 079

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize publication readiness and produce publication-grade decisions for Synthesis Design.

Prompt: Simulate a high-pressure Q&A; on Synthesis Design. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=theory-oriented reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Optimization

80. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - venue-specific optimization - Variant 080

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize novelty and produce publication-grade decisions for Search Strategy.

Prompt: Act as a senior research editor. Build a contribution architecture for Search Strategy in systematic review planning phase targeting PRISMA-aligned journal. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=methodology purist. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Planning

81. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - decision traceability - Variant 081

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize rigor and produce publication-grade decisions for Critical Appraisal.

Prompt: Rewrite the methodology around Critical Appraisal for state-of-the-art positioning for rebuttal so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ACM CSUR. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=interdisciplinary committee member. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Execution

82. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - citation-strength positioning - Variant 082

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize reproducibility and produce publication-grade decisions for Gap Identification.

Prompt: Write a Nature-style abstract for a study on Gap Identification in rapid scoping review for proposal. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=ethics compliance reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Compliance

83. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - defense-ready reasoning - Variant 083

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize clarity and produce publication-grade decisions for Synthesis Design.

Prompt: Generate a figure strategy for Synthesis Design with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Knowledge-Based Systems. Add one caption rewrite for non-expert readers. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=reproducibility auditor. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Reasoning

84. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - high-confidence statistical reporting - Variant 084

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize statistical power and produce publication-grade decisions for Search Strategy.

Prompt: Build an ethics checklist and mitigation plan for Search Strategy in systematic review planning phase. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for PRISMA-aligned journal review. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=applied domain expert. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from an applied domain expert.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Rebuttal

85. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - evidence-first writing - Variant 085

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize transferability and produce publication-grade decisions for Critical Appraisal.

Prompt: Design a publication strategy for Critical Appraisal: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ACM CSUR. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=systems performance reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Validation

86. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - hypothesis stress testing - Variant 086

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize ethical compliance and produce publication-grade decisions for Gap Identification.

Prompt: Assume Reviewer #2 challenges Gap Identification for IEEE Survey Journal. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=industry impact reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Optimization

87. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - results-to-insight translation - Variant 087

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize computational efficiency and produce publication-grade decisions for Synthesis Design.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Synthesis Design. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for Elsevier Knowledge-Based Systems. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=editor focused on clarity. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Planning

88. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - major-revision readiness - Variant 088

Objective: Select robust statistical strategy and interpretation logic. Prioritize clinical relevance and produce publication-grade decisions for Search Strategy.

Prompt: Design a statistical analysis workflow for Search Strategy with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for PRISMA-aligned journal. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=skeptical statistician. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Strategy

89. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - replication-grade documentation - Variant 089

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize publication readiness and produce publication-grade decisions for Critical Appraisal.

Prompt: Analyze a draft on Critical Appraisal and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to ACM CSUR. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=benchmarking specialist. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Analysis

90. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - reviewer-resilient argumentation - Variant 090

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize novelty and produce publication-grade decisions for Gap Identification.

Prompt: Interpret results for Gap Identification using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in IEEE Survey Journal. Add one paragraph that preempts overclaiming. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=theory-oriented reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Reasoning

91. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - ethics-by-design reporting - Variant 091

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize rigor and produce publication-grade decisions for Synthesis Design.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Synthesis Design and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for Elsevier Knowledge-Based Systems. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=methodology purist. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Rebuttal

92. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - limitations-first transparency - Variant 092

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize reproducibility and produce publication-grade decisions for Search Strategy.

Prompt: Generate a publication-ready section for Search Strategy tuned for PRISMA-aligned journal. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=interdisciplinary committee member. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Validation

93. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - cross-domain readability -

Variant 093

Objective: Pressure-test experimental design before expensive execution. Prioritize clarity and produce publication-grade decisions for Critical Appraisal.

Prompt: Audit the experimental design for Critical Appraisal. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for ACM CSUR reviewers. Include a stop/go decision protocol. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=ethics compliance reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Optimization

94. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - impact-focused communication - Variant 094

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize statistical power and produce publication-grade decisions for Gap Identification.

Prompt: Create a LaTeX writing workflow for Gap Identification: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for IEEE Survey Journal submission checks. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=reproducibility auditor. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Planning

95. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - publishable novelty

framing - Variant 095

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize transferability and produce publication-grade decisions for Synthesis Design.

Prompt: Simulate a high-pressure Q&A; on Synthesis Design. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=applied domain expert. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Strategy

96. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - practical deployment framing - Variant 096

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize ethical compliance and produce publication-grade decisions for Search Strategy.

Prompt: Act as a senior research editor. Build a contribution architecture for Search Strategy in systematic review planning phase targeting PRISMA-aligned journal. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=systems performance reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Analysis

97. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - method comparison clarity - Variant 097

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize computational efficiency and produce publication-grade decisions for Critical Appraisal.

Prompt: Rewrite the methodology around Critical Appraisal for state-of-the-art positioning for rebuttal so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ACM CSUR. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=industry impact reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Writing

98. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - editorial risk reduction - Variant 098

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize clinical relevance and produce publication-grade decisions for Gap Identification.

Prompt: Write a Nature-style abstract for a study on Gap Identification in rapid scoping review for proposal. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=editor focused on clarity. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Publication

99. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - venue-specific

optimization - Variant 099

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize publication readiness and produce publication-grade decisions for Synthesis Design.

Prompt: Generate a figure strategy for Synthesis Design with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Knowledge-Based Systems. Add one caption rewrite for non-expert readers. Project profile:

discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=skeptical statistician. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Validation

100. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - decision traceability - Variant 100

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize novelty and produce publication-grade decisions for Search Strategy.

Prompt: Build an ethics checklist and mitigation plan for Search Strategy in systematic review planning phase. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for PRISMA-aligned journal review. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=benchmarking specialist. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Optimization

101. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - citation-strength positioning - Variant 101

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize rigor and produce publication-grade decisions for Critical Appraisal.

Prompt: Design a publication strategy for Critical Appraisal: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ACM CSUR. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=theory-oriented reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Planning

102. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - defense-ready reasoning - Variant 102

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize reproducibility and produce publication-grade decisions for Gap Identification.

Prompt: Assume Reviewer #2 challenges Gap Identification for IEEE Survey Journal. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=methodology purist. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Strategy

103. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - high-confidence statistical reporting - Variant 103

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize clarity and produce publication-grade decisions for Synthesis Design.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Synthesis Design. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for Elsevier Knowledge-Based Systems. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=interdisciplinary committee member. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Analysis

104. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - evidence-first writing - Variant 104

Objective: Select robust statistical strategy and interpretation logic. Prioritize statistical power and produce publication-grade decisions for Search Strategy.

Prompt: Design a statistical analysis workflow for Search Strategy with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for PRISMA-aligned journal. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=ethics compliance reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Writing

105. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - hypothesis stress testing - Variant 105

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize transferability and produce publication-grade decisions for Critical Appraisal.

Prompt: Analyze a draft on Critical Appraisal and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to ACM CSUR. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=reproducibility auditor. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Publication

106. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - results-to-insight translation - Variant 106

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize ethical compliance and produce publication-grade decisions for Gap Identification.

Prompt: Interpret results for Gap Identification using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in IEEE Survey Journal. Add one paragraph that preempts overclaiming. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=applied domain expert. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from an applied domain expert.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Methodology

107. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - major-revision readiness - Variant 107

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize computational efficiency and produce publication-grade decisions for Synthesis Design.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Synthesis Design and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for Elsevier Knowledge-Based Systems. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=systems performance reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Execution

108. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - replication-grade documentation - Variant 108

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize clinical relevance and produce publication-grade decisions for Search Strategy.

Prompt: Generate a publication-ready section for Search Strategy tuned for PRISMA-aligned journal. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=industry impact reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Planning

109. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - reviewer-resilient argumentation - Variant 109

Objective: Pressure-test experimental design before expensive execution. Prioritize publication readiness and produce publication-grade decisions for Critical Appraisal.

Prompt: Audit the experimental design for Critical Appraisal. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for ACM CSUR reviewers. Include a stop/go decision protocol. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=editor focused on clarity. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Strategy

110. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - ethics-by-design reporting - Variant 110

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize novelty and produce publication-grade decisions for Gap Identification.

Prompt: Create a LaTeX writing workflow for Gap Identification: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for IEEE Survey Journal submission checks. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=skeptical statistician. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Analysis

111. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - limitations-first

transparency - Variant 111

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize rigor and produce publication-grade decisions for Synthesis Design.

Prompt: Simulate a high-pressure Q&A; on Synthesis Design. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=benchmarking specialist. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Writing

112. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - cross-domain readability - Variant 112

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize reproducibility and produce publication-grade decisions for Search Strategy.

Prompt: Act as a senior research editor. Build a contribution architecture for Search Strategy in systematic review planning phase targeting PRISMA-aligned journal. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=theory-oriented reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Publication

113. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - impact-focused

communication - Variant 113

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize clarity and produce publication-grade decisions for Critical Appraisal.

Prompt: Rewrite the methodology around Critical Appraisal for state-of-the-art positioning for rebuttal so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ACM CSUR. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=methodology purist. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Methodology

114. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - publishable novelty framing - Variant 114

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize statistical power and produce publication-grade decisions for Gap Identification.

Prompt: Write a Nature-style abstract for a study on Gap Identification in rapid scoping review for proposal. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=interdisciplinary committee member. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Execution

115. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - practical deployment framing - Variant 115

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize transferability and produce publication-grade decisions for Synthesis Design.

Prompt: Generate a figure strategy for Synthesis Design with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Knowledge-Based Systems. Add one caption rewrite for non-expert readers. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=ethics compliance reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Compliance

116. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - method comparison clarity - Variant 116

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize ethical compliance and produce publication-grade decisions for Search Strategy.

Prompt: Build an ethics checklist and mitigation plan for Search Strategy in systematic review planning phase. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for PRISMA-aligned journal review. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=reproducibility auditor. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Reasoning

117. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - editorial risk reduction - Variant 117

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize computational efficiency and produce publication-grade decisions for Critical Appraisal.

Prompt: Design a publication strategy for Critical Appraisal: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ACM CSUR. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=applied domain expert. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Analysis

118. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - venue-specific optimization - Variant 118

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize clinical relevance and produce publication-grade decisions for Gap Identification.

Prompt: Assume Reviewer #2 challenges Gap Identification for IEEE Survey Journal. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=systems performance reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Writing

119. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - decision traceability - Variant 119

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize publication readiness and produce publication-grade decisions for Synthesis Design.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Synthesis Design. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for Elsevier Knowledge-Based Systems. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=industry impact reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Publication

120. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - citation-strength positioning - Variant 120

Objective: Select robust statistical strategy and interpretation logic. Prioritize novelty and produce publication-grade decisions for Search Strategy.

Prompt: Design a statistical analysis workflow for Search Strategy with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for PRISMA-aligned journal. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=editor focused on clarity. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Methodology

121. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - defense-ready reasoning - Variant 121

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize rigor and produce publication-grade decisions for Critical Appraisal.

Prompt: Analyze a draft on Critical Appraisal and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to ACM CSUR. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=skeptical statistician. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Execution

122. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - high-confidence statistical reporting - Variant 122

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize reproducibility and produce publication-grade decisions for Gap Identification.

Prompt: Interpret results for Gap Identification using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in IEEE Survey Journal. Add one paragraph that preempts overclaiming. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=benchmarking specialist. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Compliance

123. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - evidence-first writing -

Variant 123

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize clarity and produce publication-grade decisions for Synthesis Design.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Synthesis Design and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for Elsevier Knowledge-Based Systems. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=theory-oriented reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Reasoning

124. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - hypothesis stress testing - Variant 124

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize statistical power and produce publication-grade decisions for Search Strategy.

Prompt: Generate a publication-ready section for Search Strategy tuned for PRISMA-aligned journal. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=methodology purist. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Rebuttal

125. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - results-to-insight translation - Variant 125

Objective: Pressure-test experimental design before expensive execution. Prioritize transferability and produce publication-grade decisions for Critical Appraisal.

Prompt: Audit the experimental design for Critical Appraisal. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for ACM CSUR reviewers. Include a stop/go decision protocol. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=interdisciplinary committee member. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Validation

126. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - major-revision readiness - Variant 126

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize ethical compliance and produce publication-grade decisions for Gap Identification.

Prompt: Create a LaTeX writing workflow for Gap Identification: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for IEEE Survey Journal submission checks. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=ethics compliance reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Publication

127. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - replication-grade

documentation - Variant 127

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize computational efficiency and produce publication-grade decisions for Synthesis Design.

Prompt: Simulate a high-pressure Q&A; on Synthesis Design. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=reproducibility auditor. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Methodology

128. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - reviewer-resilient argumentation - Variant 128

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize clinical relevance and produce publication-grade decisions for Search Strategy.

Prompt: Act as a senior research editor. Build a contribution architecture for Search Strategy in systematic review planning phase targeting PRISMA-aligned journal. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=applied domain expert. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Execution

129. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - ethics-by-design reporting -

Variant 129

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize publication readiness and produce publication-grade decisions for Critical Appraisal.

Prompt: Rewrite the methodology around Critical Appraisal for state-of-the-art positioning for rebuttal so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ACM CSUR. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=systems performance reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Compliance

130. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - limitations-first transparency - Variant 130

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize novelty and produce publication-grade decisions for Gap Identification.

Prompt: Write a Nature-style abstract for a study on Gap Identification in rapid scoping review for proposal. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=industry impact reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Reasoning

131. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - cross-domain readability - Variant 131

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize rigor and produce publication-grade decisions for Synthesis Design.

Prompt: Generate a figure strategy for Synthesis Design with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Knowledge-Based Systems. Add one caption rewrite for non-expert readers. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=editor focused on clarity. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Rebuttal

132. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - impact-focused communication - Variant 132

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize reproducibility and produce publication-grade decisions for Search Strategy.

Prompt: Build an ethics checklist and mitigation plan for Search Strategy in systematic review planning phase. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for PRISMA-aligned journal review. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=skeptical statistician. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Validation

133. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - publishable novelty framing - Variant 133

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize clarity and produce publication-grade decisions for Critical Appraisal.

Prompt: Design a publication strategy for Critical Appraisal: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ACM CSUR. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=benchmarking specialist. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Optimization

134. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - practical deployment framing - Variant 134

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize statistical power and produce publication-grade decisions for Gap Identification.

Prompt: Assume Reviewer #2 challenges Gap Identification for IEEE Survey Journal. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=theory-oriented reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Planning

135. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - method comparison clarity - Variant 135

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize transferability and produce publication-grade decisions for Synthesis Design.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Synthesis Design. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for Elsevier Knowledge-Based Systems. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=methodology purist. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Execution

136. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - editorial risk reduction - Variant 136

Objective: Select robust statistical strategy and interpretation logic. Prioritize ethical compliance and produce publication-grade decisions for Search Strategy.

Prompt: Design a statistical analysis workflow for Search Strategy with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for PRISMA-aligned journal. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=interdisciplinary committee member. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Compliance

137. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - venue-specific optimization -

Variant 137

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize computational efficiency and produce publication-grade decisions for Critical Appraisal.

Prompt: Analyze a draft on Critical Appraisal and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to ACM CSUR. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=ethics compliance reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Reasoning

138. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - decision traceability - Variant 138

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize clinical relevance and produce publication-grade decisions for Gap Identification.

Prompt: Interpret results for Gap Identification using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in IEEE Survey Journal. Add one paragraph that preempts overclaiming. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=reproducibility auditor. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Rebuttal

139. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - citation-strength positioning - Variant 139

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize publication readiness and produce publication-grade decisions for Synthesis Design.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Synthesis Design and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for Elsevier Knowledge-Based Systems. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=applied domain expert. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Validation

140. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - defense-ready reasoning - Variant 140

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize novelty and produce publication-grade decisions for Search Strategy.

Prompt: Generate a publication-ready section for Search Strategy tuned for PRISMA-aligned journal. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=systems performance reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Optimization

141. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - high-confidence statistical reporting - Variant 141

Objective: Pressure-test experimental design before expensive execution. Prioritize rigor and produce publication-grade decisions for Critical Appraisal.

Prompt: Audit the experimental design for Critical Appraisal. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for ACM CSUR reviewers. Include a stop/go decision protocol. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=industry impact reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Planning

142. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - evidence-first writing - Variant 142

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize reproducibility and produce publication-grade decisions for Gap Identification.

Prompt: Create a LaTeX writing workflow for Gap Identification: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for IEEE Survey Journal submission checks. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=editor focused on clarity. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Strategy

143. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - hypothesis stress testing - Variant 143

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize clarity and produce publication-grade decisions for Synthesis Design.

Prompt: Simulate a high-pressure Q&A; on Synthesis Design. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=skeptical statistician. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Analysis

144. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - results-to-insight translation - Variant 144

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize statistical power and produce publication-grade decisions for Search Strategy.

Prompt: Act as a senior research editor. Build a contribution architecture for Search Strategy in systematic review planning phase targeting PRISMA-aligned journal. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=benchmarking specialist. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Reasoning

145. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - major-revision readiness - Variant 145

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize transferability and produce publication-grade decisions for Critical Appraisal.

Prompt: Rewrite the methodology around Critical Appraisal for state-of-the-art positioning for rebuttal so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ACM CSUR. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=theory-oriented reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Rebuttal

146. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - replication-grade documentation - Variant 146

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize ethical compliance and produce publication-grade decisions for Gap Identification.

Prompt: Write a Nature-style abstract for a study on Gap Identification in rapid scoping review for proposal. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=methodology purist. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Validation

147. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - reviewer-resilient argumentation - Variant 147

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize computational efficiency and produce publication-grade decisions for Synthesis Design.

Prompt: Generate a figure strategy for Synthesis Design with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Knowledge-Based Systems. Add one caption rewrite for non-expert readers. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=interdisciplinary committee member. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Optimization

148. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - ethics-by-design reporting - Variant 148

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize clinical relevance and produce publication-grade decisions for Search Strategy.

Prompt: Build an ethics checklist and mitigation plan for Search Strategy in systematic review planning phase. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for PRISMA-aligned journal review. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=ethics compliance reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Planning

149. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - limitations-first transparency - Variant 149

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize publication readiness and produce publication-grade decisions for Critical Appraisal.

Prompt: Design a publication strategy for Critical Appraisal: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ACM CSUR. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=reproducibility auditor. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Strategy

150. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - cross-domain readability - Variant 150

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize novelty and produce publication-grade decisions for Gap Identification.

Prompt: Assume Reviewer #2 challenges Gap Identification for IEEE Survey Journal. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=applied domain expert. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from an applied domain expert.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Analysis

151. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - impact-focused communication - Variant 151

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize rigor and produce publication-grade decisions for Synthesis Design.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Synthesis Design. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure

that anticipates editor priorities for Elsevier Knowledge-Based Systems. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=systems performance reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Writing

152. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - publishable novelty framing - Variant 152

Objective: Select robust statistical strategy and interpretation logic. Prioritize reproducibility and produce publication-grade decisions for Search Strategy.

Prompt: Design a statistical analysis workflow for Search Strategy with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for PRISMA-aligned journal. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=industry impact reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Publication

153. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - practical deployment framing - Variant 153

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize clarity and produce publication-grade decisions for Critical Appraisal.

Prompt: Analyze a draft on Critical Appraisal and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to ACM CSUR. Project profile: discipline=computational neuroscience;

dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=editor focused on clarity. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Validation

154. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - method comparison clarity - Variant 154

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize statistical power and produce publication-grade decisions for Gap Identification.

Prompt: Interpret results for Gap Identification using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in IEEE Survey Journal. Add one paragraph that preempts overclaiming. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=skeptical statistician. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Optimization

155. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - editorial risk reduction - Variant 155

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize transferability and produce publication-grade decisions for Synthesis Design.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Synthesis Design and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for Elsevier Knowledge-Based Systems. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk

scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=benchmarking specialist. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Planning

156. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - venue-specific optimization - Variant 156

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize ethical compliance and produce publication-grade decisions for Search Strategy.

Prompt: Generate a publication-ready section for Search Strategy tuned for PRISMA-aligned journal. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=theory-oriented reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Strategy

157. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - decision traceability - Variant 157

Objective: Pressure-test experimental design before expensive execution. Prioritize computational efficiency and produce publication-grade decisions for Critical Appraisal.

Prompt: Audit the experimental design for Critical Appraisal. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for ACM CSUR reviewers. Include a stop/go decision protocol. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline};

resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=methodology purist. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Analysis

158. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - citation-strength positioning - Variant 158

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize clinical relevance and produce publication-grade decisions for Gap Identification.

Prompt: Create a LaTeX writing workflow for Gap Identification: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for IEEE Survey Journal submission checks. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=interdisciplinary committee member. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Writing

159. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - defense-ready reasoning - Variant 159

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize publication readiness and produce publication-grade decisions for Synthesis Design.

Prompt: Simulate a high-pressure Q&A; on Synthesis Design. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=ethics compliance reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Publication

160. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - high-confidence statistical reporting - Variant 160

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize novelty and produce publication-grade decisions for Search Strategy.

Prompt: Act as a senior research editor. Build a contribution architecture for Search Strategy in systematic review planning phase targeting PRISMA-aligned journal. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=reproducibility auditor. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Methodology

161. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - evidence-first writing - Variant 161

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize rigor and produce publication-grade decisions for Critical Appraisal.

Prompt: Rewrite the methodology around Critical Appraisal for state-of-the-art positioning for rebuttal so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ACM CSUR. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as

numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=applied domain expert. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Execution

162. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - hypothesis stress testing - Variant 162

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize reproducibility and produce publication-grade decisions for Gap Identification.

Prompt: Write a Nature-style abstract for a study on Gap Identification in rapid scoping review for proposal. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=systems performance reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Planning

163. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - results-to-insight translation - Variant 163

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize clarity and produce publication-grade decisions for Synthesis Design.

Prompt: Generate a figure strategy for Synthesis Design with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Knowledge-Based Systems. Add one caption rewrite for non-expert readers. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=industry impact reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Strategy

164. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - major-revision readiness - Variant 164

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize statistical power and produce publication-grade decisions for Search Strategy.

Prompt: Build an ethics checklist and mitigation plan for Search Strategy in systematic review planning phase. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for PRISMA-aligned journal review. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=editor focused on clarity. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Analysis

165. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - replication-grade documentation - Variant 165

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize transferability and produce publication-grade decisions for Critical Appraisal.

Prompt: Design a publication strategy for Critical Appraisal: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ACM CSUR. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=skeptical statistician. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Writing

166. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - reviewer-resilient argumentation - Variant 166

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize ethical compliance and produce publication-grade decisions for Gap Identification.

Prompt: Assume Reviewer #2 challenges Gap Identification for IEEE Survey Journal. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=benchmarking specialist. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Publication

167. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - ethics-by-design reporting - Variant 167

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize computational efficiency and produce publication-grade decisions for Synthesis Design.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Synthesis Design. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for Elsevier Knowledge-Based Systems. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=theory-oriented reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Methodology

168. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - limitations-first transparency - Variant 168

Objective: Select robust statistical strategy and interpretation logic. Prioritize clinical relevance and produce publication-grade decisions for Search Strategy.

Prompt: Design a statistical analysis workflow for Search Strategy with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for PRISMA-aligned journal. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=methodology purist. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Execution

169. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - cross-domain readability - Variant 169

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize publication readiness and produce publication-grade decisions for Critical Appraisal.

Prompt: Analyze a draft on Critical Appraisal and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to ACM CSUR. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=interdisciplinary committee member. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Compliance

170. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - impact-focused communication - Variant 170

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize novelty and produce publication-grade decisions for Gap Identification.

Prompt: Interpret results for Gap Identification using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in IEEE Survey Journal. Add one paragraph that preempts overclaiming. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=ethics compliance reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Reasoning

171. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - publishable novelty framing - Variant 171

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize rigor and produce publication-grade decisions for Synthesis Design.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Synthesis Design and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for Elsevier Knowledge-Based Systems. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=reproducibility auditor. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Analysis

172. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - practical deployment framing - Variant 172

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize reproducibility and produce publication-grade decisions for Search Strategy.

Prompt: Generate a publication-ready section for Search Strategy tuned for PRISMA-aligned journal. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=applied domain expert. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Writing

173. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - method comparison clarity - Variant 173

Objective: Pressure-test experimental design before expensive execution. Prioritize clarity and produce publication-grade decisions for Critical Appraisal.

Prompt: Audit the experimental design for Critical Appraisal. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for ACM CSUR reviewers. Include a stop/go decision protocol. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=systems performance reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Publication

174. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - editorial risk reduction - Variant 174

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize statistical power and produce publication-grade decisions for Gap Identification.

Prompt: Create a LaTeX writing workflow for Gap Identification: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for IEEE Survey Journal submission checks. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=industry impact reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Methodology

175. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - venue-specific optimization - Variant 175

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize transferability and produce publication-grade decisions for Synthesis Design.

Prompt: Simulate a high-pressure Q&A; on Synthesis Design. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=editor focused on clarity. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Execution

176. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - decision traceability - Variant 176

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize ethical compliance and produce publication-grade decisions for Search Strategy.

Prompt: Act as a senior research editor. Build a contribution architecture for Search Strategy in systematic review planning phase targeting PRISMA-aligned journal. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=skeptical statistician. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Compliance

177. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - citation-strength positioning - Variant 177

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize computational efficiency and produce publication-grade decisions for Critical Appraisal.

Prompt: Rewrite the methodology around Critical Appraisal for state-of-the-art positioning for rebuttal so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ACM CSUR. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=benchmarking

specialist. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Reasoning

178. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - defense-ready reasoning - Variant 178

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize clinical relevance and produce publication-grade decisions for Gap Identification.

Prompt: Write a Nature-style abstract for a study on Gap Identification in rapid scoping review for proposal. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=theory-oriented reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Rebuttal

179. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - high-confidence statistical reporting - Variant 179

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize publication readiness and produce publication-grade decisions for Synthesis Design.

Prompt: Generate a figure strategy for Synthesis Design with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Knowledge-Based Systems. Add one caption rewrite for non-expert readers. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=methodology purist. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Validation

180. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - evidence-first writing - Variant 180

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize novelty and produce publication-grade decisions for Search Strategy.

Prompt: Build an ethics checklist and mitigation plan for Search Strategy in systematic review planning phase. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for PRISMA-aligned journal review. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=interdisciplinary committee member. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Publication

181. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - hypothesis stress testing - Variant 181

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize rigor and produce publication-grade decisions for Critical Appraisal.

Prompt: Design a publication strategy for Critical Appraisal: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ACM CSUR. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=ethics compliance reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and

milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Methodology

182. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - results-to-insight translation - Variant 182

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize reproducibility and produce publication-grade decisions for Gap Identification.

Prompt: Assume Reviewer #2 challenges Gap Identification for IEEE Survey Journal. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=reproducibility auditor. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Execution

183. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - major-revision readiness - Variant 183

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize clarity and produce publication-grade decisions for Synthesis Design.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Synthesis Design. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for Elsevier Knowledge-Based Systems. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=applied domain expert. Example outcome: evidence map with risk controls, reviewer-facing

rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Compliance

184. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - replication-grade documentation - Variant 184

Objective: Select robust statistical strategy and interpretation logic. Prioritize statistical power and produce publication-grade decisions for Search Strategy.

Prompt: Design a statistical analysis workflow for Search Strategy with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for PRISMA-aligned journal. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=systems performance reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Reasoning

185. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - reviewer-resilient argumentation - Variant 185

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize transferability and produce publication-grade decisions for Critical Appraisal.

Prompt: Analyze a draft on Critical Appraisal and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to ACM CSUR. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=industry impact reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Rebuttal

186. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - ethics-by-design reporting - Variant 186

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize ethical compliance and produce publication-grade decisions for Gap Identification.

Prompt: Interpret results for Gap Identification using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in IEEE Survey Journal. Add one paragraph that preempts overclaiming. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=editor focused on clarity. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Validation

187. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - limitations-first transparency - Variant 187

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize computational efficiency and produce publication-grade decisions for Synthesis Design.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Synthesis Design and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for Elsevier Knowledge-Based Systems. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=skeptical statistician. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical

statistician.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Optimization

188. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - cross-domain readability - Variant 188

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize clinical relevance and produce publication-grade decisions for Search Strategy.

Prompt: Generate a publication-ready section for Search Strategy tuned for PRISMA-aligned journal. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=benchmarking specialist. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Planning

189. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - impact-focused communication - Variant 189

Objective: Pressure-test experimental design before expensive execution. Prioritize publication readiness and produce publication-grade decisions for Critical Appraisal.

Prompt: Audit the experimental design for Critical Appraisal. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for ACM CSUR reviewers. Include a stop/go decision protocol. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=theory-oriented reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Execution

190. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - publishable novelty framing - Variant 190

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize novelty and produce publication-grade decisions for Gap Identification.

Prompt: Create a LaTeX writing workflow for Gap Identification: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for IEEE Survey Journal submission checks. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=methodology purist. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Compliance

191. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - practical deployment framing - Variant 191

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize rigor and produce publication-grade decisions for Synthesis Design.

Prompt: Simulate a high-pressure Q&A; on Synthesis Design. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=interdisciplinary committee member. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Reasoning

192. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - method comparison clarity - Variant 192

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize reproducibility and produce publication-grade decisions for Search Strategy.

Prompt: Act as a senior research editor. Build a contribution architecture for Search Strategy in systematic review planning phase targeting PRISMA-aligned journal. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=ethics compliance reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Rebuttal

193. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - editorial risk reduction - Variant 193

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize clarity and produce publication-grade decisions for Critical Appraisal.

Prompt: Rewrite the methodology around Critical Appraisal for state-of-the-art positioning for rebuttal so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ACM CSUR. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=reproducibility auditor. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Validation

194. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - venue-specific optimization - Variant 194

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize statistical power and produce publication-grade decisions for Gap Identification.

Prompt: Write a Nature-style abstract for a study on Gap Identification in rapid scoping review for proposal. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=applied domain expert. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Optimization

195. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - decision traceability - Variant 195

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize transferability and produce publication-grade decisions for Synthesis Design.

Prompt: Generate a figure strategy for Synthesis Design with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Knowledge-Based Systems. Add one caption rewrite for non-expert readers. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=systems performance reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Planning

196. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - citation-strength positioning - Variant 196

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize ethical compliance and produce publication-grade decisions for Search Strategy.

Prompt: Build an ethics checklist and mitigation plan for Search Strategy in systematic review planning phase. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for PRISMA-aligned journal review. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=industry impact reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Strategy

197. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - defense-ready reasoning - Variant 197

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize computational efficiency and produce publication-grade decisions for Critical Appraisal.

Prompt: Design a publication strategy for Critical Appraisal: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ACM CSUR. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=editor focused on clarity. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Analysis

198. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - high-confidence statistical reporting - Variant 198

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize clinical relevance and produce publication-grade decisions for Gap Identification.

Prompt: Assume Reviewer #2 challenges Gap Identification for IEEE Survey Journal. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=skeptical statistician. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Reasoning

199. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - evidence-first writing - Variant 199

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize publication readiness and produce publication-grade decisions for Synthesis Design.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Synthesis Design. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for Elsevier Knowledge-Based Systems. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=benchmarking specialist. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Rebuttal

200. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - hypothesis stress testing - Variant 200

Objective: Select robust statistical strategy and interpretation logic. Prioritize novelty and produce publication-grade decisions for Search Strategy.

Prompt: Design a statistical analysis workflow for Search Strategy with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for PRISMA-aligned journal. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=theory-oriented reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Validation

201. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - results-to-insight translation - Variant 201

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize rigor and produce publication-grade decisions for Critical Appraisal.

Prompt: Analyze a draft on Critical Appraisal and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to ACM CSUR. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=methodology purist. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Optimization

202. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - major-revision readiness - Variant 202

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize reproducibility and produce publication-grade decisions for Gap Identification.

Prompt: Interpret results for Gap Identification using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in IEEE Survey Journal. Add one paragraph that preempts overclaiming. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=interdisciplinary committee member. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Planning

203. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - replication-grade documentation - Variant 203

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize clarity and produce publication-grade decisions for Synthesis Design.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Synthesis Design and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for Elsevier Knowledge-Based Systems. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=ethics compliance reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Strategy

204. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - reviewer-resilient argumentation - Variant 204

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize statistical power and produce publication-grade decisions for Search Strategy.

Prompt: Generate a publication-ready section for Search Strategy tuned for PRISMA-aligned journal. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=reproducibility auditor. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Analysis

205. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - ethics-by-design reporting - Variant 205

Objective: Pressure-test experimental design before expensive execution. Prioritize transferability and produce publication-grade decisions for Critical Appraisal.

Prompt: Audit the experimental design for Critical Appraisal. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for ACM CSUR reviewers. Include a stop/go decision protocol. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=applied domain expert. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from an applied domain expert.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Writing

206. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - limitations-first transparency - Variant 206

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize ethical compliance and produce publication-grade decisions for Gap Identification.

Prompt: Create a LaTeX writing workflow for Gap Identification: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for IEEE Survey Journal submission checks. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=systems performance reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Publication

207. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - cross-domain readability - Variant 207

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize computational efficiency and produce publication-grade decisions for Synthesis Design.

Prompt: Simulate a high-pressure Q&A; on Synthesis Design. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=industry impact reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Validation

208. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - impact-focused communication - Variant 208

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize clinical relevance and produce publication-grade decisions for Search Strategy.

Prompt: Act as a senior research editor. Build a contribution architecture for Search Strategy in systematic review planning phase targeting PRISMA-aligned journal. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=editor focused on clarity. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Optimization

209. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - publishable novelty framing - Variant 209

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize publication readiness and produce publication-grade decisions for Critical Appraisal.

Prompt: Rewrite the methodology around Critical Appraisal for state-of-the-art positioning for rebuttal so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ACM CSUR. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=skeptical statistician. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Planning

210. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - practical deployment framing - Variant 210

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize novelty and produce publication-grade decisions for Gap Identification.

Prompt: Write a Nature-style abstract for a study on Gap Identification in rapid scoping review for proposal. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=benchmarking specialist. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Strategy

211. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - method comparison clarity - Variant 211

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize rigor and produce publication-grade decisions for Synthesis Design.

Prompt: Generate a figure strategy for Synthesis Design with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Knowledge-Based Systems. Add one caption rewrite for non-expert readers. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=theory-oriented reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Analysis

212. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - editorial risk reduction - Variant 212

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize reproducibility and produce publication-grade decisions for Search Strategy.

Prompt: Build an ethics checklist and mitigation plan for Search Strategy in systematic review planning phase. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for PRISMA-aligned journal review. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=methodology purist. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Writing

213. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - venue-specific optimization - Variant 213

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize clarity and produce publication-grade decisions for Critical Appraisal.

Prompt: Design a publication strategy for Critical Appraisal: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ACM CSUR. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=interdisciplinary committee member. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Publication

214. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - decision traceability - Variant 214

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize statistical power and produce publication-grade decisions for Gap Identification.

Prompt: Assume Reviewer #2 challenges Gap Identification for IEEE Survey Journal. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=ethics compliance reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Methodology

215. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - citation-strength positioning - Variant 215

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize transferability and produce publication-grade decisions for Synthesis Design.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Synthesis Design. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for Elsevier Knowledge-Based Systems. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=reproducibility auditor. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Execution

216. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - defense-ready reasoning - Variant 216

Objective: Select robust statistical strategy and interpretation logic. Prioritize ethical compliance and produce publication-grade decisions for Search Strategy.

Prompt: Design a statistical analysis workflow for Search Strategy with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for PRISMA-aligned journal. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=applied domain expert. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Planning

217. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - high-confidence statistical reporting - Variant 217

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize computational efficiency and produce publication-grade decisions for Critical Appraisal.

Prompt: Analyze a draft on Critical Appraisal and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to ACM CSUR. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=systems performance reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Strategy

218. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - evidence-first writing - Variant 218

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize clinical relevance and produce publication-grade decisions for Gap Identification.

Prompt: Interpret results for Gap Identification using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in IEEE Survey Journal. Add one paragraph that preempts overclaiming. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=industry impact reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Analysis

219. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - hypothesis stress testing - Variant 219

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize publication readiness and produce publication-grade decisions for Synthesis Design.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Synthesis Design and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for Elsevier Knowledge-Based Systems. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=editor focused on clarity. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Writing

220. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - results-to-insight translation - Variant 220

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize novelty and produce publication-grade decisions for Search Strategy.

Prompt: Generate a publication-ready section for Search Strategy tuned for PRISMA-aligned journal. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=skeptical statistician. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Publication

221. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - major-revision readiness - Variant 221

Objective: Pressure-test experimental design before expensive execution. Prioritize rigor and produce publication-grade decisions for Critical Appraisal.

Prompt: Audit the experimental design for Critical Appraisal. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for ACM CSUR reviewers. Include a stop/go decision protocol. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=benchmarking specialist. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Methodology

222. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - replication-grade documentation - Variant 222

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize reproducibility and produce publication-grade decisions for Gap Identification.

Prompt: Create a LaTeX writing workflow for Gap Identification: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for IEEE Survey Journal submission checks. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=theory-oriented reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Execution

223. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - reviewer-resilient argumentation - Variant 223

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize clarity and produce publication-grade decisions for Synthesis Design.

Prompt: Simulate a high-pressure Q&A; on Synthesis Design. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=methodology purist. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Compliance

224. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - ethics-by-design reporting - Variant 224

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize statistical power and produce publication-grade decisions for Search Strategy.

Prompt: Act as a senior research editor. Build a contribution architecture for Search Strategy in systematic review planning phase targeting PRISMA-aligned journal. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=interdisciplinary committee member. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Reasoning

225. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - limitations-first transparency - Variant 225

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize transferability and produce publication-grade decisions for Critical Appraisal.

Prompt: Rewrite the methodology around Critical Appraisal for state-of-the-art positioning for rebuttal so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ACM CSUR. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=ethics compliance reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Analysis

226. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - cross-domain readability - Variant 226

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize ethical compliance and produce publication-grade decisions for Gap Identification.

Prompt: Write a Nature-style abstract for a study on Gap Identification in rapid scoping review for proposal. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=reproducibility auditor. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Writing

227. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - impact-focused communication - Variant 227

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize computational efficiency and produce publication-grade decisions for Synthesis Design.

Prompt: Generate a figure strategy for Synthesis Design with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Knowledge-Based Systems. Add one caption rewrite for non-expert readers. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=applied domain expert. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from an applied domain expert.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Publication

228. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - publishable novelty framing - Variant 228

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize clinical relevance and produce publication-grade decisions for Search Strategy.

Prompt: Build an ethics checklist and mitigation plan for Search Strategy in systematic review planning phase. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for PRISMA-aligned journal review. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=systems performance reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Methodology

229. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - practical deployment framing - Variant 229

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize publication readiness and produce publication-grade decisions for Critical Appraisal.

Prompt: Design a publication strategy for Critical Appraisal: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ACM CSUR. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=industry impact reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Execution

230. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - method comparison clarity - Variant 230

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize novelty and produce publication-grade decisions for Gap Identification.

Prompt: Assume Reviewer #2 challenges Gap Identification for IEEE Survey Journal. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=editor focused on clarity. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Compliance

231. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - editorial risk reduction - Variant 231

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize rigor and produce publication-grade decisions for Synthesis Design.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Synthesis Design. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for Elsevier Knowledge-Based Systems. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=skeptical statistician. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Reasoning

232. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - venue-specific optimization - Variant 232

Objective: Select robust statistical strategy and interpretation logic. Prioritize reproducibility and produce publication-grade decisions for Search Strategy.

Prompt: Design a statistical analysis workflow for Search Strategy with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for PRISMA-aligned journal. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=benchmarking specialist. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Rebuttal

233. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - decision traceability - Variant 233

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize clarity and produce publication-grade decisions for Critical Appraisal.

Prompt: Analyze a draft on Critical Appraisal and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to ACM CSUR. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=theory-oriented reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Validation

234. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - citation-strength positioning - Variant 234

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize statistical power and produce publication-grade decisions for Gap Identification.

Prompt: Interpret results for Gap Identification using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in IEEE Survey Journal. Add one paragraph that preempts overclaiming. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=methodology purist. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Publication

235. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - defense-ready reasoning - Variant 235

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize transferability and produce publication-grade decisions for Synthesis Design.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Synthesis Design and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for Elsevier Knowledge-Based Systems. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=interdisciplinary committee member. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Methodology

236. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - high-confidence statistical reporting - Variant 236

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize ethical compliance and produce publication-grade decisions for Search Strategy.

Prompt: Generate a publication-ready section for Search Strategy tuned for PRISMA-aligned journal. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=ethics compliance reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Execution

237. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - evidence-first writing - Variant 237

Objective: Pressure-test experimental design before expensive execution. Prioritize computational efficiency and produce publication-grade decisions for Critical Appraisal.

Prompt: Audit the experimental design for Critical Appraisal. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for ACM CSUR reviewers. Include a stop/go decision protocol. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=reproducibility auditor. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Compliance

238. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - hypothesis stress testing - Variant 238

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize clinical relevance and produce publication-grade decisions for Gap Identification.

Prompt: Create a LaTeX writing workflow for Gap Identification: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for IEEE Survey Journal submission checks. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=applied domain expert. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Reasoning

239. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - results-to-insight translation - Variant 239

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize publication readiness and produce publication-grade decisions for Synthesis Design.

Prompt: Simulate a high-pressure Q&A; on Synthesis Design. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=systems performance reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Rebuttal

240. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - major-revision readiness - Variant 240

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize novelty and produce publication-grade decisions for Search Strategy.

Prompt: Act as a senior research editor. Build a contribution architecture for Search Strategy in systematic review planning phase targeting PRISMA-aligned journal. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=industry impact reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Validation

241. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - replication-grade documentation - Variant 241

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize rigor and produce publication-grade decisions for Critical Appraisal.

Prompt: Rewrite the methodology around Critical Appraisal for state-of-the-art positioning for rebuttal so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ACM CSUR. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=editor focused on clarity. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Optimization

242. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - reviewer-resilient argumentation - Variant 242

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize reproducibility and produce publication-grade decisions for Gap Identification.

Prompt: Write a Nature-style abstract for a study on Gap Identification in rapid scoping review for proposal. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=skeptical statistician. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Planning

243. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - ethics-by-design reporting - Variant 243

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize clarity and produce publication-grade decisions for Synthesis Design.

Prompt: Generate a figure strategy for Synthesis Design with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Knowledge-Based Systems. Add one caption rewrite for non-expert readers. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=benchmarking specialist. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Execution

244. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - limitations-first transparency - Variant 244

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize statistical power and produce publication-grade decisions for Search Strategy.

Prompt: Build an ethics checklist and mitigation plan for Search Strategy in systematic review planning phase. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for PRISMA-aligned journal review. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=theory-oriented reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Compliance

245. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - cross-domain readability - Variant 245

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize transferability and produce publication-grade decisions for Critical Appraisal.

Prompt: Design a publication strategy for Critical Appraisal: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ACM CSUR. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=methodology purist. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Reasoning

246. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - impact-focused communication - Variant 246

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize ethical compliance and produce publication-grade decisions for Gap Identification.

Prompt: Assume Reviewer #2 challenges Gap Identification for IEEE Survey Journal. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=interdisciplinary committee member. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Rebuttal

247. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - publishable novelty framing - Variant 247

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize computational efficiency and produce publication-grade decisions for Synthesis Design.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Synthesis Design. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for Elsevier Knowledge-Based Systems. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=ethics compliance reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Validation

248. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - practical deployment

framing - Variant 248

Objective: Select robust statistical strategy and interpretation logic. Prioritize clinical relevance and produce publication-grade decisions for Search Strategy.

Prompt: Design a statistical analysis workflow for Search Strategy with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for PRISMA-aligned journal. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=reproducibility auditor. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Optimization

249. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - method comparison clarity - Variant 249

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize publication readiness and produce publication-grade decisions for Critical Appraisal.

Prompt: Analyze a draft on Critical Appraisal and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to ACM CSUR. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=applied domain expert. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from an applied domain expert.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Planning

250. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - editorial risk reduction - Variant 250

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize novelty and produce publication-grade decisions for Gap Identification.

Prompt: Interpret results for Gap Identification using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in IEEE Survey Journal. Add one paragraph that preempts overclaiming. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=systems performance reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Strategy

251. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - venue-specific optimization - Variant 251

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize rigor and produce publication-grade decisions for Synthesis Design.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Synthesis Design and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for Elsevier Knowledge-Based Systems. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=industry impact reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Analysis

252. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - decision traceability - Variant 252

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize reproducibility and produce publication-grade decisions for Search Strategy.

Prompt: Generate a publication-ready section for Search Strategy tuned for PRISMA-aligned journal. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=editor focused on clarity. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Reasoning

253. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - citation-strength positioning - Variant 253

Objective: Pressure-test experimental design before expensive execution. Prioritize clarity and produce publication-grade decisions for Critical Appraisal.

Prompt: Audit the experimental design for Critical Appraisal. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for ACM CSUR reviewers. Include a stop/go decision protocol. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=skeptical statistician. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Rebuttal

254. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - defense-ready reasoning - Variant 254

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize statistical power and produce publication-grade decisions for Gap Identification.

Prompt: Create a LaTeX writing workflow for Gap Identification: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for IEEE Survey Journal submission checks. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=benchmarking specialist. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Validation

255. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - high-confidence statistical reporting - Variant 255

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize transferability and produce publication-grade decisions for Synthesis Design.

Prompt: Simulate a high-pressure Q&A; on Synthesis Design. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=theory-oriented reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Optimization

256. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - evidence-first writing - Variant 256

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize ethical compliance and produce publication-grade decisions for Search Strategy.

Prompt: Act as a senior research editor. Build a contribution architecture for Search Strategy in systematic review planning phase targeting PRISMA-aligned journal. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=methodology purist. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Planning

257. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - hypothesis stress testing - Variant 257

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize computational efficiency and produce publication-grade decisions for Critical Appraisal.

Prompt: Rewrite the methodology around Critical Appraisal for state-of-the-art positioning for rebuttal so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ACM CSUR. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=interdisciplinary committee member. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Strategy

258. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - results-to-insight translation - Variant 258

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize clinical relevance and produce publication-grade decisions for Gap Identification.

Prompt: Write a Nature-style abstract for a study on Gap Identification in rapid scoping review for proposal. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=ethics compliance reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Analysis

259. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - major-revision readiness - Variant 259

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize publication readiness and produce publication-grade decisions for Synthesis Design.

Prompt: Generate a figure strategy for Synthesis Design with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Knowledge-Based Systems. Add one caption rewrite for non-expert readers. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=reproducibility auditor. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Writing

260. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - replication-grade documentation - Variant 260

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize novelty and produce publication-grade decisions for Search Strategy.

Prompt: Build an ethics checklist and mitigation plan for Search Strategy in systematic review planning phase. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for PRISMA-aligned journal review. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=applied domain expert. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Publication

261. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - reviewer-resilient argumentation - Variant 261

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize rigor and produce publication-grade decisions for Critical Appraisal.

Prompt: Design a publication strategy for Critical Appraisal: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ACM CSUR. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=systems performance reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Validation

262. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - ethics-by-design reporting - Variant 262

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize reproducibility and produce publication-grade decisions for Gap Identification.

Prompt: Assume Reviewer #2 challenges Gap Identification for IEEE Survey Journal. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=industry impact reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Optimization

263. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - limitations-first transparency - Variant 263

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize clarity and produce publication-grade decisions for Synthesis Design.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Synthesis Design. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for Elsevier Knowledge-Based Systems. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=editor focused on clarity. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Planning

264. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - cross-domain readability - Variant 264

Objective: Select robust statistical strategy and interpretation logic. Prioritize statistical power and produce publication-grade decisions for Search Strategy.

Prompt: Design a statistical analysis workflow for Search Strategy with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for PRISMA-aligned journal. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=skeptical statistician. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Strategy

265. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - impact-focused communication - Variant 265

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize transferability and produce publication-grade decisions for Critical Appraisal.

Prompt: Analyze a draft on Critical Appraisal and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to ACM CSUR. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=benchmarking specialist. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Analysis

266. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - publishable novelty framing - Variant 266

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize ethical compliance and produce publication-grade decisions for Gap Identification.

Prompt: Interpret results for Gap Identification using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in IEEE Survey Journal. Add one paragraph that preempts overclaiming. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=theory-oriented reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Writing

267. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - practical deployment framing - Variant 267

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize computational efficiency and produce publication-grade decisions for Synthesis Design.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Synthesis Design and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for Elsevier Knowledge-Based Systems. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=methodology purist. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Publication

268. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - method comparison clarity - Variant 268

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize clinical relevance and produce publication-grade decisions for Search Strategy.

Prompt: Generate a publication-ready section for Search Strategy tuned for PRISMA-aligned journal. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=interdisciplinary committee member. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Methodology

269. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - editorial risk reduction - Variant 269

Objective: Pressure-test experimental design before expensive execution. Prioritize publication readiness and produce publication-grade decisions for Critical Appraisal.

Prompt: Audit the experimental design for Critical Appraisal. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for ACM CSUR reviewers. Include a stop/go decision protocol. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=ethics compliance reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Execution

270. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - venue-specific optimization - Variant 270

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize novelty and produce publication-grade decisions for Gap Identification.

Prompt: Create a LaTeX writing workflow for Gap Identification: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for IEEE Survey Journal submission checks. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=reproducibility auditor. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Planning

271. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - decision traceability - Variant 271

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize rigor and produce publication-grade decisions for Synthesis Design.

Prompt: Simulate a high-pressure Q&A; on Synthesis Design. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=applied domain expert. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Strategy

272. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - citation-strength positioning - Variant 272

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize reproducibility and produce publication-grade decisions for Search Strategy.

Prompt: Act as a senior research editor. Build a contribution architecture for Search Strategy in systematic review planning phase targeting PRISMA-aligned journal. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=systems performance reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Analysis

273. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - defense-ready reasoning - Variant 273

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize clarity and produce publication-grade decisions for Critical Appraisal.

Prompt: Rewrite the methodology around Critical Appraisal for state-of-the-art positioning for rebuttal so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ACM CSUR. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=industry impact reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Writing

274. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - high-confidence statistical reporting - Variant 274

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize statistical power and produce publication-grade decisions for Gap Identification.

Prompt: Write a Nature-style abstract for a study on Gap Identification in rapid scoping review for proposal. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=editor focused on clarity. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Publication

275. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - evidence-first writing - Variant 275

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize transferability and produce publication-grade decisions for Synthesis Design.

Prompt: Generate a figure strategy for Synthesis Design with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Knowledge-Based Systems. Add one caption rewrite for non-expert readers. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=skeptical statistician. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Methodology

276. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - hypothesis stress testing - Variant 276

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize ethical compliance and produce publication-grade decisions for Search Strategy.

Prompt: Build an ethics checklist and mitigation plan for Search Strategy in systematic review planning phase. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for PRISMA-aligned journal review. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=benchmarking specialist. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Execution

277. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - results-to-insight translation - Variant 277

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize computational efficiency and produce publication-grade decisions for Critical Appraisal.

Prompt: Design a publication strategy for Critical Appraisal: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ACM CSUR. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=theory-oriented reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Compliance

278. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - major-revision readiness - Variant 278

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize clinical relevance and produce publication-grade decisions for Gap Identification.

Prompt: Assume Reviewer #2 challenges Gap Identification for IEEE Survey Journal. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and

fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=methodology purist. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Reasoning

279. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - replication-grade documentation - Variant 279

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize publication readiness and produce publication-grade decisions for Synthesis Design.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Synthesis Design. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for Elsevier Knowledge-Based Systems. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=interdisciplinary committee member. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Analysis

280. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - reviewer-resilient argumentation - Variant 280

Objective: Select robust statistical strategy and interpretation logic. Prioritize novelty and produce publication-grade decisions for Search Strategy.

Prompt: Design a statistical analysis workflow for Search Strategy with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for PRISMA-aligned journal. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=ethics compliance reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Writing

281. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - ethics-by-design reporting - Variant 281

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize rigor and produce publication-grade decisions for Critical Appraisal.

Prompt: Analyze a draft on Critical Appraisal and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to ACM CSUR. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=reproducibility auditor. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Publication

282. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - limitations-first transparency - Variant 282

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize reproducibility and produce publication-grade decisions for Gap Identification.

Prompt: Interpret results for Gap Identification using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in IEEE Survey Journal. Add one paragraph that preempts overclaiming. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=applied domain expert. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Methodology

283. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - cross-domain readability - Variant 283

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize clarity and produce publication-grade decisions for Synthesis Design.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Synthesis Design and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for Elsevier Knowledge-Based Systems. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=systems performance reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Execution

284. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - impact-focused communication - Variant 284

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize statistical power and produce publication-grade decisions for Search Strategy.

Prompt: Generate a publication-ready section for Search Strategy tuned for PRISMA-aligned journal. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=industry impact reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Compliance

285. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - publishable novelty framing - Variant 285

Objective: Pressure-test experimental design before expensive execution. Prioritize transferability and produce publication-grade decisions for Critical Appraisal.

Prompt: Audit the experimental design for Critical Appraisal. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for ACM CSUR reviewers. Include a stop/go decision protocol. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=editor focused on clarity. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Reasoning

286. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - practical deployment framing - Variant 286

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize ethical compliance and produce publication-grade decisions for Gap Identification.

Prompt: Create a LaTeX writing workflow for Gap Identification: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for IEEE Survey Journal submission checks. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=skeptical statistician. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Rebuttal

287. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - method comparison clarity - Variant 287

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize computational efficiency and produce publication-grade decisions for Synthesis Design.

Prompt: Simulate a high-pressure Q&A; on Synthesis Design. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=benchmarking specialist. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Validation

288. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - editorial risk reduction - Variant 288

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize clinical relevance and produce publication-grade decisions for Search Strategy.

Prompt: Act as a senior research editor. Build a contribution architecture for Search Strategy in systematic review planning phase targeting PRISMA-aligned journal. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project

profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=theory-oriented reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Publication

289. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - venue-specific optimization - Variant 289

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize publication readiness and produce publication-grade decisions for Critical Appraisal.

Prompt: Rewrite the methodology around Critical Appraisal for state-of-the-art positioning for rebuttal so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for ACM CSUR. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=methodology purist. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Methodology

290. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - decision traceability - Variant 290

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize novelty and produce publication-grade decisions for Gap Identification.

Prompt: Write a Nature-style abstract for a study on Gap Identification in rapid scoping review for proposal. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=interdisciplinary

committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=interdisciplinary committee member. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Execution

291. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - citation-strength positioning - Variant 291

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize rigor and produce publication-grade decisions for Synthesis Design.

Prompt: Generate a figure strategy for Synthesis Design with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Knowledge-Based Systems. Add one caption rewrite for non-expert readers. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=ethics compliance reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Compliance

292. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - defense-ready reasoning - Variant 292

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize reproducibility and produce publication-grade decisions for Search Strategy.

Prompt: Build an ethics checklist and mitigation plan for Search Strategy in systematic review planning phase. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for PRISMA-aligned journal review. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=reproducibility auditor. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Reasoning

293. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - high-confidence statistical reporting - Variant 293

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize clarity and produce publication-grade decisions for Critical Appraisal.

Prompt: Design a publication strategy for Critical Appraisal: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for ACM CSUR. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=applied domain expert. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Rebuttal

294. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - evidence-first writing - Variant 294

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize statistical power and produce publication-grade decisions for Gap Identification.

Prompt: Assume Reviewer #2 challenges Gap Identification for IEEE Survey Journal. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and

fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=systems performance reviewer. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Validation

295. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - hypothesis stress testing - Variant 295

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize transferability and produce publication-grade decisions for Synthesis Design.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Synthesis Design. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for Elsevier Knowledge-Based Systems. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=industry impact reviewer. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Optimization

296. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - results-to-insight translation - Variant 296

Objective: Select robust statistical strategy and interpretation logic. Prioritize ethical compliance and produce publication-grade decisions for Search Strategy.

Prompt: Design a statistical analysis workflow for Search Strategy with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for PRISMA-aligned journal. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=editor focused on clarity. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Planning

297. Create a Review Synthesis That Cites Correctly: related-work architecture for state-of-the-art positioning for rebuttal (ACM CSUR) - major-revision readiness - Variant 297

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize computational efficiency and produce publication-grade decisions for Critical Appraisal.

Prompt: Analyze a draft on Critical Appraisal and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to ACM CSUR. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=ACM CSUR; reviewer persona=skeptical statistician. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Critical Appraisal.

Difficulty: Beginner

Category: Execution

298. Detect Citation Gaps with Confidence: gap extraction engine for rapid scoping review for proposal (IEEE Survey Journal) - replication-grade documentation - Variant 298

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize clinical relevance and produce publication-grade decisions for Gap Identification.

Prompt: Interpret results for Gap Identification using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in IEEE Survey Journal. Add one paragraph that preempts overclaiming. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=benchmarking specialist;

output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE Survey Journal; reviewer persona=benchmarking specialist. Example outcome: thematic synthesis with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready thematic synthesis that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Gap Identification.

Difficulty: Intermediate

Category: Compliance

299. Turn Papers into Research Direction: evidence grading workflow for chapter-2 thesis drafting cycle (Elsevier Knowledge-Based Systems) - reviewer-resilient argumentation - Variant 299

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize publication readiness and produce publication-grade decisions for Synthesis Design.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Synthesis Design and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for Elsevier Knowledge-Based Systems. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Knowledge-Based Systems; reviewer persona=theory-oriented reviewer. Example outcome: citation logic matrix with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready citation logic matrix that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Synthesis Design.

Difficulty: Advanced

Category: Reasoning

300. Build a Literature Map That Finds Gaps: query-to-synthesis workflow for systematic review planning phase (PRISMA-aligned journal) - ethics-by-design reporting - Variant 300

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize novelty and produce publication-grade decisions for Search Strategy.

Prompt: Generate a publication-ready section for Search Strategy tuned for PRISMA-aligned journal. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement;

timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=PRISMA-aligned journal; reviewer persona=methodology purist. Example outcome: evidence map with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready evidence map that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Search Strategy.

Difficulty: Expert

Category: Rebuttal

Summary

Applying these 300 workflows systematically improves decision quality, writing clarity, and publication readiness across research cycles.